

Organizational Accessibility: From Self-Maintenance to Evolvability

A Dynamical Framework for Cumulative Emergence

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Abstract

Why can organization accumulate rather than repeatedly requiring rediscovery from scratch? We propose that the relevant object is **organizational accessibility**: the finite-horizon distribution of organizational states that can follow from what already exists. The claim is not merely that present state affects future state, which is true of any dynamical system. It is that realized organization can become part of the causal conditions determining which successor organizations are effectively accessible.

We construct the account in four steps: a baseline self-maintenance threshold followed by three distinct mechanisms. First, an open production network can collectively replace the processes that constitute it, yielding an organization reproduction number, R_{org} . Below one, organization introduced at low abundance decays; above one, it can establish. Second (**Mechanism I — hysteretic retention**), a seed-dependent feedback can separate establishment from maintenance. In the worked model, a rare process C cannot invade the resident A – B organization below $J_{\text{inv}} = 1.0667$, yet once established the expanded A – B – C organization persists down to $J_{\text{fold}} = 0.5978$. At identical external conditions, realized state and basin occupancy can therefore change **what can remain**. Third (**Mechanism II — inherited adaptive extension**), reproduction with inheritance makes an already viable organization the starting architecture for subsequent variation rather than requiring viability to be reconstructed de novo. Continued reproduction also supplies a finite exploratory opportunity before extinction. Near a persistence boundary, a declining lineage can therefore combine many remaining variant-generation trials with a comparatively small adaptive displacement required to restore positive growth. This connects organizational accessibility to evolutionary rescue and eco-evolutionary feedback. Fourth (**Mechanism III — evolvability**), the process that generates candidate successors can itself vary: the distribution of variants, their generation rate, or both may change and be differentially retained.

We formalize the common object with a finite-horizon accessibility kernel $\mathcal{A}_\tau(S \mid \theta, z, E, Q)$, where θ denotes organizational architecture, z its current dynamical state, E the external environment, and $Q = (q, \nu)$ the variant-generating process. For novel successors, exploratory opportunity

$$\Omega_\tau = \mathbb{E} \left[\int_t^{t+\tau} N(s) \nu(s) ds \right]$$

combines with the conditional variant distribution $q(\theta' \mid \theta)$ and establishment probability π_{est} to determine a successful-successor intensity. The three mechanisms are mathematically distinct: Mechanism I changes effective persistence accessibility through basin structure, Mechanism II changes the inherited starting architecture and the amount of search available, and Mechanism III changes Q itself.

The framework implies neither a force toward complexity nor monotonically increasing accessibility. History may widen, narrow, fragment, or collapse the set of viable successors. The general claim is instead: **existence changes what existence can follow**. Cumulative organization is the special case in which enough previous viability is retained that it becomes infrastructure for subsequent viability, so later viable organization need not repeatedly be reconstructed from scratch.

1 The problem: not survival of parts, but continuation of organization

Organized systems persist despite continuous turnover. Molecules decay, cells die, organisms replace cells, institutions replace members, machines replace components, and computational processes terminate and restart. Persistence therefore need not mean persistence of the same material parts. It can instead mean persistence of a causal organization: a network of processes whose activity repeatedly produces the processes required for the network to continue.

This shifts the explanatory target. The first question is not merely why a component survives, but how a collection of locally acting processes can collectively cause its own continuation. Once such continuation is possible, a second question appears:

How can realized organization become part of the causal conditions determining the accessibility of its own successors?

This formulation is deliberately stronger than the trivial statement that present state affects future state, but weaker than claiming that every realized state rewrites the underlying laws of transition. The claim developed here is about **effective organizational accessibility**. History can matter because an established state occupies a different basin, because descendants inherit an already viable architecture, because reproductive history changes the number of exploratory trials available, because realized organization changes its environment, or because the variant-generating process itself changes.

The argument proceeds in four steps. Sections 2–3 derive a baseline reproduction threshold for self-maintaining organization. Sections 5–6 establish **Mechanism I — hysteretic retention**: a newly established feedback can alter the conditions required for its own continued existence. Section 7 develops **Mechanism II — inherited adaptive extension**: once organization is reproduced, successor search begins from inherited viable architecture and is supplied by a finite, history-dependent exploratory opportunity. Section 8 develops **Mechanism III — evolvability**: the process generating candidate successors can itself vary and be retained.

The construction sits near several established traditions. Autocatalytic-set theory studies networks in which reactions are collectively sustained by catalysts produced within the network (Hordijk, Hein, et al., 2010; Hordijk, Steel, et al., 2012). Jain and Krishna showed how autocatalytic structure can emerge and reorganize in evolving catalytic networks (Jain and Krishna, 1998; Jain and Krishna, 2001). Peng, Linderoth and Baum formalized seed-dependent autocatalytic systems that can remain active after a transient seed and can form hierarchical structures (Peng et al., 2022). Evolutionary-rescue theory studies how populations facing environmental deterioration can avoid extinction through adaptation before decline is complete (Bell, 2017; Ferrière and Legendre, 2013), while pathogen-emergence theory has shown that subcritical transmission chains become longer as a reproduction number approaches one, increasing cumulative opportunities for adaptive change (Antia et al., 2003; Woolhouse et al., 2005).

The present contribution is narrower than a general theory of complexity. We use an explicit resource balance to derive a reproduction threshold, show how one added feedback produces a

backward-bifurcation geometry, distinguish that mechanism from inheritance-mediated adaptive extension near a persistence boundary, and connect both to variation in the process that generates successor organizations. The aim is not to prove that complexity increases. It is to identify measurable ways in which organizational history can condition what organization is accessible next.

1.1 Organizational accessibility

To keep architecture distinct from instantaneous state, let

$$\theta = \text{organizational architecture}, \quad z = \text{instantaneous dynamical state}.$$

For example, in the models below θ may specify the available process types and interaction structure, while z contains their current abundances and the resource state. Let E denote the external environment. Let

$$Q = (q, \nu)$$

denote the process generating organizational variants, where $q(\theta' \mid \theta)$ is the conditional distribution of candidate successor architectures given a variation event and ν is the rate of such events per active realization.

For a measurable region S of the joint architecture–state space and a finite horizon τ , define the **organizational-accessibility kernel**

$$\boxed{\mathcal{A}_\tau(S \mid \theta, z, E, Q) = \Pr((\Theta_{t+\tau}, Z_{t+\tau}) \in S \mid \Theta_t = \theta, Z_t = z, E, Q).} \quad (1)$$

The exact state space and horizon must be specified in any real application; E and Q may denote fixed values or specified exogenous trajectories over that horizon. In a deterministic model, this kernel may collapse onto one or several outcomes determined by architecture, dynamical state, and basin occupancy. In a stochastic or evolutionary system, it is a non-trivial distribution over successor organizational states.

Equation (1) is a transition kernel at a specified horizon. If the empirical question is instead whether the process **ever reaches** a viable successor before time τ , the appropriate object is a finite-horizon hitting probability. That distinction is useful because persistence, first passage, and novel-successor generation need not have the same mathematical form.

For novel-successor search, define **exploratory opportunity** as

$$\boxed{\Omega_\tau = \mathbb{E} \left[\int_t^{t+\tau} N(s) \nu(s) ds \right],} \quad (2)$$

where $N(s)$ is the number of active realizations. In a deterministic or mean-field treatment, the expectation may be omitted. Ω_τ is the expected number of relevant candidate-generation events over the horizon; it is not the standing population and not an independent primitive of the system, because it is generated by (θ, z, E, Q) and by prior history.

A generated candidate still need not establish. Let

$$\pi_{\text{est}}(\theta' \mid \theta, z, E) \in [0, 1]$$

denote the probability that candidate architecture θ' establishes or otherwise enters the realized trajectory when generated from current architecture θ and state z in environment E . For a measurable architecture set $U \subseteq \Theta$, a minimal intensity of successfully established novel successors in U is then

$$\Lambda_\tau(U \mid \theta, z, E, Q) = \Omega_\tau \int_U q(\theta' \mid \theta) \pi_{\text{est}}(\theta' \mid \theta, z, E) d\theta'. \quad (3)$$

Under an independent Poisson approximation,

$$\Pr(\text{at least one successful novel successor in } U) \approx 1 - e^{-\Lambda_\tau(U)}. \quad (4)$$

Equation (3) gives a useful conceptual decomposition:

$$\text{successful-successor accessibility} \sim \text{opportunities} \times \text{what is generated} \times \text{what can establish}.$$

The factorization is not claimed to be exact for every system. Its value is that it separates quantities that can often be perturbed or estimated independently.

The three mechanisms developed below enter this framework through different causal channels:

- **Mechanism I — hysteretic retention.** At fixed external conditions, realized dynamical state and basin occupancy can determine whether an already established organization remains. Realization changes **what can remain**.
- **Mechanism II — inherited adaptive extension.** Inheritance makes θ the base point from which $q(\theta' \mid \theta)$ is sampled, while reproductive history changes the $N(t)$ contribution to Ω_τ . Realization changes **where search begins and how much search occurs**.
- **Mechanism III — evolvability.** Organization can alter $Q = (q, \nu)$ itself. Realization changes **how candidate successors are generated**.

The environment E remains conceptually distinct. In eco-evolutionary systems, realized organization may also alter a future environment E_{t+1} , thereby changing later accessibility through an additional feedback. The SARS-CoV-2 illustration in Section 7.3 uses this route, but environmental modification is not required for the framework.

None of these mechanisms need increase complexity, adaptability, or persistence. Accessibility may widen, narrow, shift, fragment, or collapse. The most general statement is descriptive:

$$\text{Existence changes what existence can follow.}$$

Cumulative emergence is a more specific regime. It occurs when historical changes in accessibility carry forward enough causal structure for viability that later viable organization can be reached without reconstructing prior viability de novo. In that case,

$$\text{previous viability can become infrastructure for subsequent viability.}$$

This motivates the working definition used throughout the paper:

A **cumulative organizational ratchet** occurs when realized organization changes subsequent effective accessibility in a way that carries forward enough previously achieved viability that later viable organization need not be rediscovered de novo.

2 A resource-limited self-maintaining production network

Let $x_i \geq 0$ denote the abundance of executing process type i , and let $r \geq 0$ denote activated substrate. The nonnegative matrix B contains local production relations: B_{ij} is the rate at

which process j produces process i per unit substrate and producer abundance. Each new unit consumes one unit of substrate. Existing processes are lost at rate $d > 0$. The environment supplies activated substrate at rate J , while unused substrate is lost at rate ℓr . The model is

$$\dot{x} = rBx - dx, \quad (5)$$

$$\dot{r} = J - \ell r - r\mathbf{1}^T Bx. \quad (6)$$

The same production term enters positively in the process balance and negatively in the substrate balance. Hence this is a replacement model rather than a model in which producers are immortal. Defining

$$T = r + \sum_i x_i,$$

we obtain

$$\dot{T} = J - \ell r - d \sum_i x_i \leq J - \min(\ell, d)T.$$

Thus total activated material remains bounded, and when $J = 0$ the active organization ultimately disappears. The organization is causally closed only in the limited sense that its processes can regenerate one another; energetically and materially it remains open.

This distinction is important. The model is not a derivation from full chemical thermodynamics. A real molecular implementation would require stoichiometric, kinetic and free-energy constraints to be specified separately. Thermodynamic work on molecular replicators makes clear that replication and selection carry energetic requirements rather than escaping them (Kolchinsky, 2025).

3 A reproduction number for organization

With no active organization, the substrate converges to

$$r_0 = \frac{J}{\ell}.$$

Near this organization-free equilibrium, the process dynamics linearize to

$$\dot{x} = (r_0 B - dI)x.$$

Let $\rho(B)$ denote the spectral radius of B . The leading growth rate is

$$s = r_0 \rho(B) - d,$$

which motivates the dimensionless organization reproduction number

$$R_{\text{org}} = \frac{J\rho(B)}{\ell d}. \quad (7)$$

If $R_{\text{org}} < 1$, sufficiently small organization decays. If $R_{\text{org}} > 1$, it can grow from low abundance. For irreducible B , let p be the positive Perron eigenvector normalized by $\mathbf{1}^T p = 1$ and let $\lambda = \rho(B)$. The positive equilibrium is

$$r^* = \frac{d}{\lambda}, \quad x^* = Np, \quad N = \frac{J - \ell d/\lambda}{d},$$

which exists precisely when $R_{\text{org}} > 1$.

The same calculation can be written in next-generation-matrix form (Driessche and Watmough, 2002). At the organization-free equilibrium,

$$F = r_0 B, \quad V = dI,$$

so

$$K = FV^{-1} = \frac{r_0}{d} B, \quad \rho(K) = \frac{J\rho(B)}{\ell d} = R_{\text{org}}.$$

The vocabulary is borrowed from infectious-disease dynamics, but the interpretation is generic: R_{org} asks whether one generation of executing processes collectively produces more than one replacement generation before being lost.

For the two-process cycle

$$B = \begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix},$$

we have $\rho(B) = \sqrt{\alpha\beta}$. Neither process need reproduce itself directly. A closed return path across multiple process types can be enough. If the return path is broken so that the matrix is acyclic, the spectral radius falls to zero. This isolates the causal object of interest: not activity alone, but a route from present activity back to renewed productive capacity.

The positive equilibrium of the base model appears continuously at $R_{\text{org}} = 1$:

$$N = \frac{J}{d} \left(1 - \frac{1}{R_{\text{org}}} \right).$$

This is the expected forward transcritical geometry. In the base model, establishment from rarity and loss of the positive equilibrium meet at the same threshold. Section 6 shows how a seed-dependent feedback can separate establishment from maintenance and thereby create hysteresis. Section 7 asks a different question while retaining the same basic threshold: how much exploratory opportunity remains, and how large an adaptive change is required, when an established organization is driven back toward and below $R_{\text{org}} = 1$.

4 An innovation must pay its opportunity cost

A new process is not automatically an improvement. To make this explicit, begin with $A \rightarrow B$ and $B \rightarrow A$ at unit coefficient. Let A divert a fraction u of its production capacity from B to a new process C , while C contributes back to production of A with coefficient v :

$$B(u, v) = \begin{pmatrix} 0 & 1 & v \\ 1 - u & 0 & 0 \\ u & 0 & 0 \end{pmatrix}, \quad 0 < u < 1, \ v \geq 0.$$

The dominant eigenvalue is

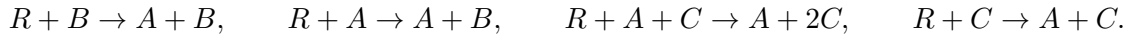
$$\lambda(u, v) = \sqrt{1 - u + uv} = \sqrt{1 + u(v - 1)}.$$

Hence the new process lowers the external supply required for maintenance only when $v > 1$. If $v < 1$, the diversion to C reduces collective replacement capacity. The model therefore contains no rule that rewards novelty, diversity, or complexity. An addition must causally return enough productive benefit to offset its cost.

This distinction matters for the later argument. None of the mechanisms below is generated by a preference for larger networks. Mechanism I changes the persistence component of accessibility because a new feedback loop alters the persistence geometry (Sections 5–6). Mechanism II instead changes the inherited starting point and the number of exploratory trials available before loss (Section 7). Mechanism III changes the distribution from which candidate successors are generated (Section 8).

5 A seed-dependent extension: history enters the state

A fixed linear production matrix does not capture an innovation that becomes self-maintaining only after it is present. We therefore consider a nonlinear extension with reactions represented by



The corresponding normalized equations are

$$\dot{a} = rb + vrc - a, \tag{8}$$

$$\dot{b} = ra - b, \tag{9}$$

$$\dot{c} = urac - c, \tag{10}$$

$$\dot{r} = J - r - (rb + vrc + ra + urac). \tag{11}$$

All production consumes substrate, and $\dot{T} = J - T$ for $T = r + a + b + c$.

The A – B subsystem can exist without C . Process C , however, reproduces only when both A and C are already present, while C contributes back to A . A one-off seed can therefore activate a loop that was dynamically inaccessible from $c = 0$.

For $u = 30$ and $v = 2$, numerical experiments give the following controlled comparison. Start the A – B system at its equilibrium with $J = 1.2$. At time $t = 50$, introduce 10^{-4} of C while subtracting the same amount from resource. At $t = 150$, lower supply to $J = 0.9$. With the $C \rightarrow A$ return path intact, A , B , and C approach a positive equilibrium. If C is never seeded, the organization disappears after the supply reduction. If C is seeded but its return effect on A is removed ($v = 0$), the organization also disappears. Thus the persistent difference is carried by the newly established feedback architecture, not by continuing external supply of C .

The seed need not be deliberately introduced. One may add a rare local reaction $R + A \rightarrow A + C$ and treat its occurrence stochastically. Once a rare event creates C , the ordinary deterministic dynamics can amplify it. This does not explain why such a reaction must exist in nature; it shows only that an external evaluator that recognizes and installs the “better” organization is not logically required.

6 Mechanism I — hysteretic retention: changing what can remain

The key result is that the threshold for establishing C is not the same as the threshold for maintaining the C -containing state.

For $c > 0$, the equilibrium conditions of Eq. (11) imply

$$a = \frac{1}{ur}, \quad b = \frac{1}{u}, \quad c = \frac{1-r^2}{uvr^2}, \quad 0 < r < 1.$$

The associated resource supply is

$$J(r) = r + \frac{1/r + 1 + (1/r^2 - 1)/v}{u}. \quad (12)$$

The fold of this positive equilibrium branch satisfies

$$\frac{dJ}{dr} = 0 \iff ur^3 - r - \frac{2}{v} = 0.$$

For $u = 30$ and $v = 2$,

$$r_{\text{fold}} \approx 0.356236, \quad J_{\text{fold}} \approx 0.597806.$$

Now examine invasion from the resident A - B state. For $J > 1$, that equilibrium has

$$r^* = 1, \quad a^* = b^* = \frac{J-1}{2}.$$

At very small c ,

$$\dot{c} \approx (ur^*a^* - 1)c,$$

so the type-reproduction number of C is

$$R_C = ur^*a^* = u \frac{J-1}{2}.$$

The invasion threshold $R_C = 1$ is therefore

$$J_{\text{inv}} = 1 + \frac{2}{u} = 1.066667. \quad (13)$$

This is the same logic used for type-specific invasion numbers in heterogeneous epidemic systems (Heesterbeek and Roberts, 2007).

The striking fact is

$$J_{\text{fold}} < 1 < J_{\text{inv}}.$$

The positive C -containing equilibrium branch extends backward from the invasion point to much lower values of J , turning at the fold. This is the characteristic geometry called a backward bifurcation in epidemiological models with sub-threshold endemic equilibria (Kribs-Zaleta and Velasco-Hernández, 2000; Driessche and Watmough, 2002).

The stable states can be summarized as follows:

Supply range	Stable unexpanded state	Stable expanded state
$J < J_{\text{fold}}$	organization-free	no
$J_{\text{fold}} < J < 1$	organization-free	A - B - C
$1 < J < J_{\text{inv}}$	A - B	A - B - C
$J > J_{\text{inv}}$	A - B unstable to rare C	A - B - C

Thus, at $J = 0.9$, an unexpanded system decays to the organization-free state while an already established A - B - C organization persists. At $J = 1.02$, a small C introduced into the

A – B state cannot invade, but an already established A – B – C state can persist. In both cases the same current environment permits different long-term outcomes depending on history. This is hysteresis.

In the accessibility formalism of Section 1.1, the effect belongs in the **persistence component of $\mathcal{A}$** , not in the external environment E itself. The environment can be numerically identical in the two histories, yet the probability mass that $\mathcal{A}$ assigns to continued occupancy of the expanded organization differs because the realized feedback architecture has altered the system’s persistence geometry.

An organizational innovation exhibits **Mechanism I — hysteretic retention** when the conditions required to establish it are stricter than the conditions required to maintain it once established.

The backward bifurcation is not synonymous with organizational accessibility; it is one explicit dynamical mechanism by which accessibility becomes history-dependent. The new loop changes its own persistence conditions. Once it exists, simply reversing the environmental change that permitted its establishment does not necessarily undo it. This is stronger than saying that positive feedback stabilizes a system. It says that realized organization changes the retention component of future accessibility: **what can remain depends on what has already been realized**.

7 Mechanism II — inherited adaptive extension: changing where search begins

Mechanism I changes what can remain by altering the retention geometry of an already realized organization. Mechanism II is different. It does not require bistability, a fold, or a newly closed feedback loop. It arises whenever a reproducing organization generates descendants from its current structure rather than from the organization-free state.

Two consequences follow.

First, inheritance changes the **base point** of subsequent search. Variants are generated around an already viable architecture θ , not around the organization-free architecture.

Second, reproduction changes the **number of draws** made from the local variation process before the lineage disappears. This is the exploratory-opportunity term Ω introduced in Eq. (2).

Together these effects create **inheritance-mediated adaptive extension**: what has already survived changes both the base point of future variation and the amount of exploration available before persistence is lost. During environmental deterioration, the same ingredients can additionally produce evolutionary rescue.

7.1 Exploratory opportunity near a reproduction threshold

Consider a simple branching process with mean reproduction number $R < 1$. Let N_g be the expected number of active realizations in generation g , beginning from N_0 founders. Then

$$E[N_g] = N_0 R^g.$$

The expected **cumulative number of realizations** before extinction is therefore

$$E[C] = \sum_{g=0}^{\infty} E[N_g] = \frac{N_0}{1 - R}, \quad R < 1. \quad (14)$$

For one founder, this reduces to the familiar $1/(1 - R)$ result. Importantly, Eq. (14) is a statement about cumulative lineage size, not the instantaneous standing population.

Now let ν denote the expected number of relevant candidate-variation events per realization. Under the simplest independent approximation, the expected exploratory opportunity generated before extinction is

$$\Omega \approx \nu E[C] = \frac{\nu N_0}{1 - R}. \quad (15)$$

Equation (15) is the corresponding deterministic/mean-field approximation to Eq. (2). Thus, for fixed N_0 and ν , a lineage driven only slightly below its persistence threshold can still generate many more candidate variants before disappearing than a lineage driven far below it. For example, one founder gives $E[C] = 2$ at $R = 0.5$, 10 at $R = 0.9$, and 100 at $R = 0.99$.

The divergence as $R \rightarrow 1^-$ is an idealized branching-process result. Finite populations, finite observation windows, competition, resource limitation, changing environments, and correlations among descendants will truncate it. The relevant general quantity is not $1/(1 - R)$ itself but Ω : the integrated number of opportunities available for generating candidate successors before the current organization is lost.

This also clarifies how pre-establishment and post-establishment situations differ. Before establishment, a rare organization typically has small N_0 , so even a favourable near-threshold value of R may yield only a limited search. After a period of supracritical reproduction, deterioration may begin from a much larger N_0 . The combination of an inherited viable structure and a large initial population can make Ω orders of magnitude larger than during the original emergence.

Standing variation present when deterioration begins provides an additional source of candidates, but it should not be folded into Ω_τ . Exploratory opportunity counts **new candidate-generation events**. Pre-existing candidate diversity contributes separately to the successful-successor intensity. A useful decomposition is

$$\Lambda_\tau(U) = \Lambda_{\text{standing}}(U) + \Omega_\tau \int_U q(\theta' | \theta) \pi_{\text{est}}(\theta' | \theta, z, E) d\theta', \quad (16)$$

where $\Lambda_{\text{standing}}(U)$ denotes the expected successful-establishment contribution, over the same horizon, of candidate variants already present at the onset of deterioration. Candidates generated after that onset belong only to the $\Omega_\tau q$ term, avoiding double counting. This keeps **standing variation** conceptually distinct from **new exploratory opportunity generated during decline**.

7.2 Near threshold, the adaptive step required for rescue can be small

Now suppose an established organization experiences environmental deterioration so that its resident reproduction number falls just below one:

$$R_{\text{res}} = 1 - \varepsilon, \quad 0 < \varepsilon \ll 1.$$

A variant with relative reproductive advantage s has approximately

$$R_{\text{var}} = (1 - \varepsilon)(1 + s).$$

The variant grows when

$$(1 - \varepsilon)(1 + s) > 1,$$

or

$$s > \frac{\varepsilon}{1 - \varepsilon} \approx \varepsilon \quad (\varepsilon \text{ small}). \quad (17)$$

A more general local form makes the geometry explicit when θ admits a differentiable local parameterization. Let $R(\theta, E)$ denote the continuation number of architecture θ in environment E , with $R(\theta, E) = 1 - \varepsilon$. For a nearby variant $\theta' = \theta + \delta\theta$,

$$R(\theta + \delta\theta, E) \approx 1 - \varepsilon + \nabla_{\theta} R(\theta, E) \cdot \delta\theta.$$

Positive growth is locally restored whenever

$$\boxed{\nabla_{\theta} R(\theta, E) \cdot \delta\theta > \varepsilon.} \quad (18)$$

This identifies a geometric opportunity, not a guarantee. Accessible variants may not include a favourable direction; the local gradient may be small; adaptation may require crossing a fitness valley; or decline may be too rapid for a suitable candidate to arise and establish.

The vicinity of the threshold is therefore special for two independent reasons.

First, the adaptive displacement required to restore positive growth is small.

Second, for fixed N_0 and variant-generation rate ν , exploratory opportunity Ω_{τ} is large because decline is slow.

These two effects multiply in the novel-successor intensity. A rescue-capable architecture region $U_{\text{rescue}} \subseteq \Theta$ is more likely to be reached when

$$\Lambda(U_{\text{rescue}}) = \Omega \int_{U_{\text{rescue}}} q(\theta' | \theta) \pi_{\text{est}}(\theta' | \theta, z, E) d\theta'$$

is large. Near threshold, Ω increases while the set of variants capable of restoring $R > 1$ can also expand because the required advantage is smaller.

This is the logic of **evolutionary rescue**: decline does not imply instantaneous extinction. It leaves a finite interval during which already existing diversity and newly generated variants may contain or produce a successor whose growth rate is positive in the changed environment (Bell, 2017). Rescue is not guaranteed. It depends on the initial abundance, the variation kernel q , standing variation, the rate of new variant production, and the establishment filter π_{est} , as well as on how quickly and how far the environment has deteriorated.

Schematically,

deterioration $\rightarrow R_{\text{res}} < 1 \rightarrow$ finite $\Omega_{\tau} \rightarrow$ candidate successor $\rightarrow R_{\text{var}} > 1 \rightarrow$ renewed expansion.

Unlike Mechanism I, no bistable branch is required. Mechanism II is therefore not a second kind of hysteresis. It is **inheritance-mediated adaptive extension under finite exploratory opportunity**.

7.3 Illustration: SARS-CoV-2 and an eco-evolutionary feedback

Pathogen evolution makes this mechanism unusually intuitive.

At the start of the SARS-CoV-2 pandemic, the human population was largely immunologically naive. Viral circulation then changed that environment by generating infection-derived

immunity, while vaccination added further immune pressure. The ecological consequence of prior viral success therefore altered the subsequent fitness landscape.

As population immunity accumulated, lineages with antigenic novelty or immune-escape properties could acquire a reproductive advantage even when their success did not arise solely from increased intrinsic transmissibility (Carabelli et al., 2023). In the language of the present framework, prior circulation did several things at once:

1. it created a very large population of already human-adapted viruses;
2. it generated standing genetic diversity and continued exploratory opportunity through further replication;
3. it changed the environment E by altering host immunity;
4. it changed which parts of successor space had high retention or establishment probability

π_{est} .

A lineage approaching its effective persistence boundary could therefore be replaced by a nearby descendant that restored $R_e > 1$ under the new immune environment.

This is an eco-evolutionary feedback: realized organization changes the environment that subsequently changes the accessibility of its descendants (Ferrière and Legendre, 2013). It is offered only as an illustration of the general mechanism. The toy model of Sections 2–6 is not a mechanistic model of SARS-CoV-2 evolution.

7.4 The inheritance corollary: search does not restart from zero

Exploratory opportunity is only half of Mechanism II. The other half is the location from which exploration begins.

Suppose the first appearance of a viable organization requires a difficult sequence of changes:

$$0 \longrightarrow \theta.$$

Once θ is reproduced with sufficient fidelity, later variants are generated conditionally on that realized organization:

$$\theta \longrightarrow \theta + \delta\theta.$$

The second search is therefore not a repeat of the first. Capabilities already embodied in θ are typically inherited unless subsequent variation disrupts them. In a pathogen, descendants of a successfully transmitting lineage usually begin with the molecular and physiological machinery that already permits host-cell entry, replication, shedding, and transmission. In the A – B – C example, descendants of the established architecture begin with the return path already present unless a variant removes it.

The important claim is therefore not that every descendant preserves every previous function. It is that the distribution $q(\theta' \mid \theta)$ is conditioned on the realized organization. Its support is centred on transformations of θ , not on independent redraws from an organization-free state.

This is the inheritance asymmetry:

successful previous organization becomes the starting point for the next search.

The original establishment problem may have required crossing a large region of organizational state space. Subsequent adaptive change can be local because previous solutions are carried into the initial condition of the next search.

In terms of organizational accessibility, inheritance changes the conditional distribution of successors even if the underlying mutation or variation rules are unchanged:

$$q(\theta' \mid \theta_1) \neq q(\theta' \mid \theta_0)$$

simply because $\theta_1 \neq \theta_0$.

This should be distinguished from evolvability. Inheritance changes the **conditioning state** of the variation process. Evolvability, developed in Section 8, changes the **variation operator itself**.

7.5 Organizational accessibility unifies the mechanisms

Mechanisms I and II are not two mathematical forms of the same bifurcation and should not share the label *hysteresis*. They are two ways in which realized organization changes the general accessibility kernel defined in Eq. (1). Equation (3) quantifies only the novel-successor search component relevant especially to Mechanisms II and III.

Mechanism I — hysteretic retention. The realized organization changes the persistence component of $\mathcal{A}$. The same external environment can retain an organization that could not establish there from rarity. What changes is **what can remain**.

Mechanism II — inherited adaptive extension. The realized organization changes the conditioning state θ from which variants are generated, and its reproductive history changes exploratory opportunity Ω_τ . What changes is **where search begins and how much local search occurs**.

A third mechanism will alter the variation process Q itself, through q , the event rate ν , or both.

Mechanism	Component of accessibility affected	What changes	Representative formal object
I — hysteretic retention	persistence mass of $\mathcal{A}_\tau$	what can remain	$J_{\text{fold}} < J_{\text{inv}}$
II — inherited adaptive extension	θ and Ω_τ	where search begins and how many trials occur	$q(\theta' \mid \theta)$ and $\Omega_\tau \simeq \nu N_0 / (1 - R)$
III — evolvability	variation process $Q_m = (q_m, \nu_m)$	how candidate successors are generated	q_m and/or ν_m

The mechanisms can compound. A feedback established through Mechanism I may preserve organization under conditions that would otherwise eliminate it. Continued reproduction can then provide the inherited base point and exploratory opportunity required for Mechanism II. A successful successor becomes a new base point for subsequent accessibility:

$$\theta_t \rightarrow \mathcal{A}_t \rightarrow \theta_{t+1} \rightarrow \mathcal{A}_{t+1} \rightarrow \dots$$

No teleology is implied. The system does not estimate ε , seek a rescuing value of s , or attempt to maximize $\mathcal{A}$. Variation is generated by the ordinary processes represented by Q and Ω ; the environment and current organization determine which variants establish and how rapidly lineages decline or expand. Near threshold, the consequences of small differences in reproductive performance become large because the dynamical filter changes, not because the system acquires a purpose.

Nor must accessibility widen. A realized organization can narrow future possibilities, trap itself on a local configuration, or fail to generate a rescuing successor before exploratory opportunity is exhausted. The ratchet relevant to cumulative emergence is therefore not an automatic tendency toward improvement. It is the historical carrying-forward of organization into the conditions from which later organization is generated.

8 Mechanism III — evolvability: changing how successor search is generated

Mechanisms I and II change effective organizational accessibility without requiring the variant-generating process itself to change. Mechanism I acts through state-dependent persistence and basin structure. Mechanism II acts through inherited starting architecture and history-dependent exploratory opportunity. A further recursion becomes possible when the **generator of candidate successors itself varies**.

Let m denote a heritable or otherwise retainable modifier of the variation process. Write

$$Q_m = (q_m, \nu_m),$$

where $q_m(\theta' | \theta)$ is the conditional distribution of candidate successor architectures and ν_m is their generation rate per active realization. A modifier may therefore change the **shape** of variation, the **rate** of variation, or both. Examples include mutation rate, recombination, developmental variability, exploratory connectivity, modularity, robustness, genotype–phenotype mapping, the probability that new feedback paths form, and mechanisms that preserve useful substructure while allowing other parts to vary.

The modifier changes exploratory opportunity through

$$\Omega_{m,\tau} = \mathbb{E} \left[\int_t^{t+\tau} N_m(s) \nu_m(s) ds \right],$$

and hence the novel-successor intensity

$$\Lambda_{m,\tau}(U) = \Omega_{m,\tau} \int_U q_m(\theta' | \theta) \pi_{\text{est}}(\theta' | \theta, z, E) d\theta'.$$

This decomposition separates two causal routes that can otherwise look similar. Mechanism II can increase exploratory opportunity because establishment and subsequent history increase $N(t)$ or prolong the time during which the lineage remains extant. Mechanism III can change exploratory opportunity because the modifier changes ν_m , and can independently change the distribution of candidates through q_m .

To connect successor accessibility to persistence consequences, let $\Phi(\theta)$ be a persistence-relevant quantity: for example the range of environments over which an organization remains stable, expected persistence under an environment distribution, or a multidimensional measure combining replacement capacity with resource cost. An accessibility-consistent innovation yield over horizon τ is

$$Y_\tau(m | \theta, z, E) = \Omega_{m,\tau} \int [\Phi(\theta') - \Phi(\theta)]_+ q_m(\theta' | \theta) \pi_{\text{est}}(\theta' | \theta, z, E) d\theta' - c_\tau(m). \quad (19)$$

Here $c_\tau(m)$ is the direct cost of maintaining or expressing modifier m over the same horizon. Equation (19) is not a biological law. It is a bookkeeping expression that places the amount

of search, the distribution of candidates, their probability of establishment, their persistence consequences, and the direct cost of the variation machinery in one common formalism.

If a modifier that changes q_m , ν_m , or both is retained together with the successful organizations it helps generate, another feedback becomes possible:

variation-generating architecture $\rightarrow$ changed successor accessibility $\rightarrow$ differential persistence or reproduction

The system can then change not only its current organization but the process that generates distributions of future organization.

This is closely related to **evolvability**: the capacity to generate heritable, selectable phenotypic variation (G. P. Wagner and Altenberg, 1996; Gerhart and Kirschner, 2007). Theory and experiments show that mutation rate, recombination, robustness, modularity, and genotype–phenotype architecture can affect evolvability. Explicit models have demonstrated conditions under which evolvability can itself be indirectly selected (Earl and Deem, 2004).

The qualification is essential. Selection for future adaptive capacity is generally indirect and may be weaker than selection on immediate performance (Díaz Arenas and Cooper, 2013; A. Wagner, 2022). A trait can affect evolvability without having been selected *because* of that effect (Pélabon et al., 2025). Increased variation can be destructive; robustness can inhibit useful change as well as protect function; modularity can constrain as well as facilitate; feedback can preserve maladaptive organization.

Mechanism III therefore does not mean that systems automatically become better at adapting. The narrower claim is:

Self-maintaining systems can, when variant-generating mechanisms themselves vary and remain coupled to persistence or reproduction, evolve changes in the rate or distribution by which successor organizations become accessible.

The resulting hierarchy is:

- **Level 1:** processes produce processes.
- **Level 2:** a network collectively maintains its organization, governed here by R_{org} .
- **Level 3 — Mechanism I:** realized feedback changes persistence accessibility through basin structure and hysteresis.
- **Level 4 — Mechanism II:** inheritance changes the starting architecture θ , while reproductive history changes the $N(t)$ contribution to exploratory opportunity Ω_τ .
- **Level 5 — Mechanism III:** the organization changes the variant-generating process $Q_m = (q_m, \nu_m)$.
- **Level 6:** those variation-generating mechanisms themselves vary and are retained differentially.

At the upper levels, the system can change the process by which distributions of its own future states are generated. This is a plausible route to cumulative evolvability, not a demonstrated consequence of the lower-level model. Modifier dynamics must be modelled and tested explicitly before any claim of evolved evolvability is warranted.

9 Empirical signatures of the three mechanisms

The framework is useful only if its mechanisms can be distinguished empirically. The first mechanism yields spectral signatures near bifurcation thresholds; the second yields a statistical

dependence of rescue on exploratory opportunity and distance from threshold; the third requires comparison of variant-generating operators.

9.1 Spectral signatures of the baseline and hysteretic thresholds

Near a stable equilibrium z^* , add weak stochastic forcing and linearize:

$$\dot{\delta z} = A(J)\delta z + \sigma\xi(t).$$

Projecting onto the slow eigenmode with eigenvalue $\lambda_{\text{slow}}(J) < 0$ yields an Ornstein–Uhlenbeck approximation. Under standard assumptions,

$$\text{Var}(\delta z) \propto \frac{1}{|\lambda_{\text{slow}}|}, \quad \rho(\Delta t) \approx 1 - |\lambda_{\text{slow}}|\Delta t.$$

Thus, as the dominant restoring eigenvalue approaches zero, recovery slows, variance increases, and lagged autocorrelation rises: the standard critical-slowness picture (Scheffer et al., 2009; Dakos et al., 2012).

For the baseline transcritical self-maintenance threshold of Section 3,

$$|\lambda_{\text{slow}}| \propto |R_{\text{org}} - 1|, \quad \text{Var} \propto |R_{\text{org}} - 1|^{-1}.$$

For the Mechanism-I C -containing branch approaching its hysteretic-loss fold from above,

$$|\lambda_{\text{slow}}| \propto \sqrt{J - J_{\text{fold}}}.$$

Numerical evaluation of the stable branch gives a fitted exponent of approximately 0.515, consistent with the generic square-root scaling of a saddle-node. Hence

$$\text{Var} \propto (J - J_{\text{fold}})^{-1/2}.$$

Two qualifications are important. First, the stable branch must be used for the return-to-equilibrium interpretation; the opposite side of the fold is unstable. Second, $1/\sqrt{\Delta J}$ diverges less strongly than $1/\Delta J$ as $\Delta J \rightarrow 0$. The fold therefore does not imply a stronger variance divergence at the same numerical distance from threshold. What differs is the local scaling law and the geometry of approach to the transition.

9.2 Mechanism II: rescue depends on exploratory opportunity and distance from threshold

Mechanism II predicts a different signature. For matched systems beginning decline with the same N_0 and variant-generation rate ν , Eq. (15) gives

$$\Omega \approx \frac{\nu N_0}{1 - R_{\text{res}}}$$

for a simple subcritical branching approximation. At the same time, Eq. (17) shows that the minimum relative reproductive advantage required for rescue is

$$s_{\text{min}} = \frac{1 - R_{\text{res}}}{R_{\text{res}}}.$$

Pushing the resident further below threshold therefore acts in two directions simultaneously: it reduces the number of exploratory trials available before extinction and increases the adaptive displacement required for any one trial to succeed.

The resulting prediction is not that a “standing population” scales as $1/(1 - R)$. It is that, conditional on comparable initial abundance and variation machinery, **rescue probability should decline as exploratory opportunity falls and the required adaptive step grows**. Systems with identical current R_{res} but different histories may also differ if one enters decline with larger N_0 , more standing variation, or a different q .

A direct test would therefore factorially manipulate:

1. distance below the persistence threshold;
2. abundance N_0 at the onset of deterioration;
3. standing variation present at that moment;
4. the rate or distribution of newly generated variants.

If recovery depends only on the current environment and not on these history-dependent sources of accessibility, Mechanism II is not supported.

9.3 Mechanism III: accessibility changes when the variation operator changes

Mechanism III requires a different experiment again. Compare otherwise matched organizations with modifiers m_1 and m_2 that differ in q_m , ν_m , or both, while controlling as far as possible for direct present-day effects.

The relevant observations are not simply whether one lineage adapts faster, but whether the modifiers measurably change:

- the distribution of candidate successors generated;
- the probability that generated candidates occupy persistence-improving regions of state space;
- the number of candidate-generation events per unit reproductive exposure;
- and, crucially, whether the modifier is differentially retained because of those consequences.

The final requirement is what distinguishes **selection on evolvability** from a trait that merely happens to alter evolvability as a side effect.

All of these predictions should be treated cautiously. Empirical early-warning signals depend on noise structure, sampling, nonstationarity, rates of parameter change, finite population size, and observability of the relevant state variables. The equations predict signatures of the present model and its simple stochastic extensions; they do not establish that any particular biological, social, technological, or artificial system is governed by the same local normal forms.

10 What the framework explains — and what it does not

The construction supports six claims.

1. **Collective self-maintenance is possible without a central controller.** Local production processes can jointly replace the processes that execute the network.
2. **Self-maintenance has a threshold.** The base model has a reproduction number R_{org} that separates decay from establishment at low abundance.
3. **Realized feedback can change its own retention conditions.** A new loop can produce bistability and hysteresis, so that an organization persists under conditions in which it could not establish from rarity. This is Mechanism I.

4. **Inheritance and reproductive history can change successor accessibility without hysteresis.** Once an organization has been realized, descendants are generated around that already viable structure, and the number of exploratory trials available before loss depends on abundance, reproduction, and variant-generation rate. Near a persistence boundary, the required adaptive displacement may be small while exploratory opportunity remains substantial. This is Mechanism II.
5. **History can change effective organizational accessibility, not merely the currently occupied state.** A realized successor can change basin occupancy, inherited architecture θ , exploratory opportunity Ω , the future environment, or the variant-generating process Q . Consequently, the accessibility encountered later in the lineage may differ because of the organization previously realized.
6. **There is a coherent route to evolvability feedback.** If variant-generating mechanisms themselves vary and remain coupled to the persistence or reproduction of the organizations they help produce, selection can act on changes in q_m , ν_m , or related innovation machinery. This is Mechanism III.

The framework does **not** show that:

- complexity must increase monotonically;
- accessibility must widen over time;
- every added process or connection is useful;
- every persistent organization is beneficial by an external criterion;
- more variation, connectivity, or feedback is always advantageous;
- proximity to $R = 1$ guarantees evolutionary rescue;
- cumulative lineage size in Eq. (14) is the same thing as instantaneous standing population size;
- inheritance preserves every previously acquired function;
- Mechanism II is a form of hysteresis;
- evolvability is always selected because of its effects on future variation;
- the present toy reactions are realized in any particular organism, ecosystem, institution, robot, or AI system;
- the number of process types is a sufficient measure of biological or informational complexity.

The same formalism allows simplification, loss, trapping, and extinction. A fast self-reproducing competitor can displace a multi-process network. An inherited architecture can constrain future search. A lineage pushed far below threshold may still fail to recover despite a large prior population if its remaining exploratory opportunity and accessible distribution of variant effects are insufficient for rescue. A lineage can exhaust its exploratory opportunity without finding a rescuing successor. A variation modifier can raise the production of harmful variants.

The framework therefore supplies mechanisms for **historical transformation of organizational accessibility**. Cumulative emergence is one possible regime of that transformation, not its inevitable direction.

11 A research programme for organizational accessibility

The framework becomes scientifically useful when its components are measured separately.

For a candidate real system, one should identify:

- a) the organizational architecture θ : the process types, interactions, or other causal relations whose continuation defines the organization;
- b) the resources consumed in their replacement;
- c) the directed causal effects by which one process increases the future abundance or execution of another;
- d) the reproduction, replacement, or invasion threshold of the existing organization;
- e) a candidate innovation and its opportunity cost;
- f) whether the innovation creates a return path into the organization that produces or maintains it;
- g) whether establishment and maintenance thresholds differ, indicating Mechanism I;
- h) whether current persistence under identical external conditions depends on historical seeding or initial conditions;
- i) the dynamical state $z(t)$ and, where relevant, the active population or number of realizations $N(t)$ through time;
- j) the rate $\nu(t)$ at which relevant candidate successors are generated, allowing estimation of exploratory opportunity

$$\Omega_\tau = \int_t^{t+\tau} N(s)\nu(s) ds;$$

- k) the distribution $q(\theta' | \theta)$ of candidate successors around the current organization;
- l) the establishment filter $\pi_{\text{est}}(\theta' | \theta, E)$ for generated candidates;
- m) whether recovery from deterioration depends on the inherited starting state, abundance at deterioration onset, standing variation, and Ω , as predicted by Mechanism II;
- n) whether variation-generating mechanisms differ among persistent lineages or organizations;
- o) whether such modifiers alter q , ν , or both;
- p) whether those modifiers are retained because of the successor organizations they make accessible rather than because of unrelated direct benefits.

The decisive experiments differ by mechanism.

For **Mechanism I**, intervene on return paths. Introduce the same candidate process in otherwise matched systems, preserve its resource cost, and selectively break the proposed feedback from consequence to renewed production. If the difference between establishment and maintenance remains unchanged, the proposed loop was not causal.

For **Mechanism II**, manipulate accessibility without changing the retention architecture. Matched systems can be driven below threshold with different N_0 , different amounts of standing variation, or different prior reproductive histories while holding the current environment fixed. The prediction is that recovery depends on the resulting Ω and on the distribution of local variants, not only on current R_{res} .

A particularly informative experiment would compare the accessibility of a target viable organization from two starting points:

$$0 \longrightarrow \theta'$$

versus

$$\theta \longrightarrow \theta',$$

where θ already contains part of the causal organization required for viability. The inheritance claim predicts that the second transition can be substantially more accessible when prior viable structure is preserved.

For **Mechanism III**, compare architectures differing in their variant-generating operator while controlling direct present-day performance. Measure not merely the number of variants produced but the resulting distribution over organizational state space and the fraction surviving the establishment filter.

The central empirical object is therefore not “complexity” in the abstract. It is the causal decomposition of accessibility:

starting organization $\times$ exploratory opportunity $\times$ variation distribution $\times$ retention dynamics.

The factorization is conceptual rather than universally multiplicative, but it identifies quantities that can be perturbed and measured separately.

12 Discussion: from persistence to the historical transformation of accessibility

Darwinian natural selection explains differential propagation of heritable variation. The present framework addresses a complementary question: **what determines the organizational variation and persistent states that are effectively accessible for selection and dynamics to act upon?** The answer proposed here is historical. Realized organization can become part of the causal conditions determining what can remain, where successor search begins, how much search occurs, and how candidate successors are generated.

The distinction from ordinary state dependence matters. A fixed dynamical law may of course give different futures from different current states. The present theory does not need to claim that every organizational transition rewrites that law. Its claim is instead that organizational history can systematically alter **effective accessibility** through several mechanistically distinguishable channels. Hysteretic basin occupancy can retain a state at fixed external conditions. Inheritance can make previously viable architecture the base point of later variation. Reproductive history can alter exploratory opportunity. Eco-evolutionary feedback can alter future environment. Evolvability can change the generator Q itself.

Mechanism I provides the most explicit dynamical demonstration. In the seed-dependent model, the conditions for establishment and maintenance separate: $J_{\text{fold}} < J_{\text{inv}}$. The same external environment can therefore support different persistent outcomes depending on which basin history has placed the system in. This is genuine hysteresis and should remain the only mechanism in the paper described by that term.

Mechanism II is different. Once a viable architecture is inherited, successor variation is conditioned on that architecture rather than drawn independently from an organization-free state. Continued activity also supplies a finite exploratory opportunity Ω_τ . Near a persistence boundary, the same lineage may simultaneously decline slowly enough to generate many

remaining trials and require only a small favourable displacement to restore positive growth. That combination is the setting of evolutionary rescue, but the deeper inheritance result does not depend on rescue: **successful previous organization becomes the starting point for the next search.**

Mechanism III takes the recursion one level further. If organizations differ in $Q_m = (q_m, \nu_m)$, then they differ not merely in present performance but in the distribution and rate of future variants they generate. Selection can then, under restricted conditions, act indirectly on properties of the search process itself. Equation (19) makes this connection explicit by combining exploratory opportunity, candidate distribution, establishment, persistence consequences, and the cost of the modifier.

These mechanisms do not imply progress. A feedback can preserve a maladaptive state. Inheritance can trap a lineage near a local configuration. Exploratory opportunity generates harmful as well as useful candidates. A changed variation operator can reduce rather than increase access to viable successors. Evolution may simplify organization or terminate lineages. The general statement is therefore not that existence makes subsequent existence easier. It is:

Existence changes what existence can follow.

The cumulative case requires additional conditions. If organization retains causal structure needed for viability, if that structure is inherited with sufficient fidelity, and if subsequent variation can modify it without reconstructing it from zero, then previous viability can become infrastructure for later viability:

previous viability can become infrastructure for subsequent viability.

This provides a precise sense in which organizational history can accumulate without invoking a force toward complexity. What accumulates need not be complexity as such; it is **causal structure that conditions later accessibility.**

The relationship to Darwinian selection can then be stated simply. Selection does not operate on all logically imaginable forms. It acts on variants that are generated, encountered, and able to establish under particular ecological and dynamical conditions. The accessibility framework asks why that presented distribution has the structure it does. Previous organization may have altered the inherited base point, the amount of search, the establishment landscape, the environment, or the variation generator itself. In that sense:

Selection acts within an accessibility structure that previous organization may itself have helped create.

The recursion is therefore better written with the full conditioning state than as a claim that a mathematical kernel is literally rewritten at every step:

$$(\theta_t, z_t, E_t, Q_t) \longrightarrow \mathcal{A}_t \longrightarrow (\theta_{t+1}, z_{t+1}, E_{t+1}, Q_{t+1}) \longrightarrow \mathcal{A}_{t+1} \longrightarrow \dots$$

The important historical possibility is that

$$\mathcal{A}_{t+1} \neq \mathcal{A}_t$$

because one or more of the causal conditions determining accessibility were changed by what happened in the intervening transition. Only Mechanism III requires the generator Q itself to change; Mechanisms I and II can transform effective accessibility through state, inheritance, and history while the underlying variation rules remain unchanged.

This suggests a restricted use of the term *ratchet*:

A cumulative organizational ratchet occurs when realized organization changes subsequent effective accessibility in a way that carries forward enough previously achieved viability that later viable organization need not be rediscovered de novo.

Mechanism I can carry viability forward by making an established state harder to lose than it was to establish. Mechanism II can carry it forward because descendants start from inherited viable architecture and retain a finite, history-dependent opportunity to explore nearby alternatives. Mechanism III can additionally carry forward properties of the process by which alternatives are generated.

The deepest empirical question is therefore not whether systems tend toward complexity, but:

When, and by what mechanisms, does realized organization become part of the causal conditions that determine future organizational accessibility?

The present models provide one constructive answer. They do not establish a universal law.

13 Conclusion

This paper develops a dynamical account of **organizational accessibility**: the finite-horizon accessibility of successor organizational states conditional on organizational architecture θ , dynamical state z , external environment E , and variant-generating process $Q = (q, \nu)$.

The argument has four layers.

First, an open network of local production processes can collectively cause its own continuation. Its establishment is governed by the reproduction threshold R_{org} . This baseline determines whether organization can grow from low abundance at all.

Second, **Mechanism I — hysteretic retention** shows how realized state can change what remains accessible at fixed external conditions. In the worked example, a seed-dependent innovation invades only above $J_{\text{inv}} = 1.0667$ but, once established, persists down to $J_{\text{fold}} = 0.5978$. The resulting backward bifurcation creates bistability and history-dependent basin occupancy.

Third, **Mechanism II — inherited adaptive extension** shows how realization changes where successor search begins and how much search can occur. Reproduction with inheritance makes already viable architecture the base point of later variation. Reproductive history generates exploratory opportunity

$$\Omega_\tau = \mathbb{E} \left[\int_t^{t+\tau} N(s) \nu(s) ds \right].$$

For a simple subcritical branching process, cumulative lineage size per founder scales as $1/(1 - R)$ as $R \rightarrow 1^-$; this is a statement about cumulative reproductive opportunity, not instantaneous standing population. Near the same boundary, the adaptive displacement required to restore positive growth may be small. Together these effects create the conditions for evolutionary rescue without requiring hysteresis.

Fourth, **Mechanism III — evolvability** extends the recursion to the variant-generating process itself. If organizations differ in $Q_m = (q_m, \nu_m)$ and those differences remain coupled to descendant persistence or reproduction, then selection can alter how candidate successors are generated. The search process becomes an evolvable component of organizational accessibility.

The common formal object is

$$\mathcal{A}_\tau(S \mid \theta, z, E, Q) = \Pr((\Theta_{t+\tau}, Z_{t+\tau}) \in S \mid \Theta_t = \theta, Z_t = z, E, Q).$$

For novel successors, the corresponding successful-successor intensity is

$$\Lambda_{\tau}(U) = \Omega_{\tau} \int_U q(\theta' \mid \theta) \pi_{\text{est}}(\theta' \mid \theta, z, E) d\theta'.$$

This decomposition distinguishes **how many opportunities occur, what variants are generated**, and **which generated variants can establish**. Inheritance determines the architecture from which that distribution is generated; hysteresis determines whether already-realized organization remains accessible; evolvability changes the generator itself.

The framework does not imply that accessibility increases, that complexity accumulates monotonically, or that evolution possesses foresight. History may narrow accessibility, preserve liabilities, constrain search, or end in simplification and extinction.

The general claim is instead:

Existence changes what existence can follow.

Under conditions that preserve enough causal structure for viability, the cumulative special case is:

What persists becomes part of the starting conditions for what can persist next.

Equivalently, previous viability can become infrastructure for subsequent viability. Cumulative emergence is therefore not fundamentally the accumulation of complexity. It is a historical regime in which realized organization conditions future accessibility while carrying forward enough of what already worked that later viability need not repeatedly be discovered from zero.

Selection then operates within an accessibility structure that previous organization may itself have helped create. That is the proposed dynamical basis of cumulative organization.

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